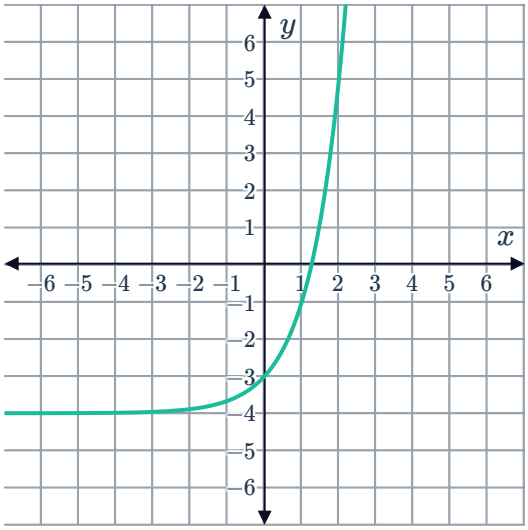
Transformations of exponential functions

1 $y = 2^x - 7$

2

a Translate the graph downwards by 4 units.

b



3

a 0 **b** 5 **c** 0 **d** $y = 5$

4

a $(0, 1)$ **b** -1 **c** $y = 0$ **d** $y = -2$

5 $g(x) = 2^x - 3$

6

a $g(x) = f(x + 4)$. A translation of 4 units to the left.

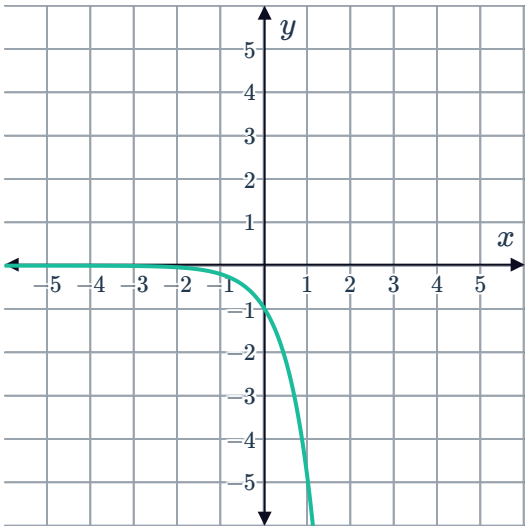
b $g(x) = 3^{x+4}$

7

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a A reflection about the x-axis.

b



8

a 4

b A reflection about the x-axis.

c $y = 4 \times 2^x$

9 $y = 3 \times 2^x$

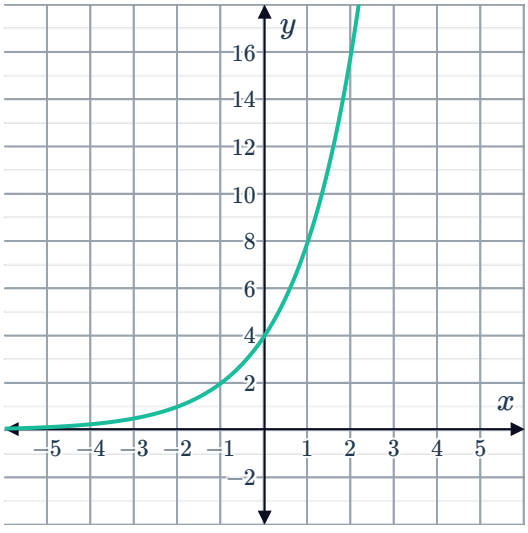
10

a $(0, 4)$

b No

c ∞

d



11

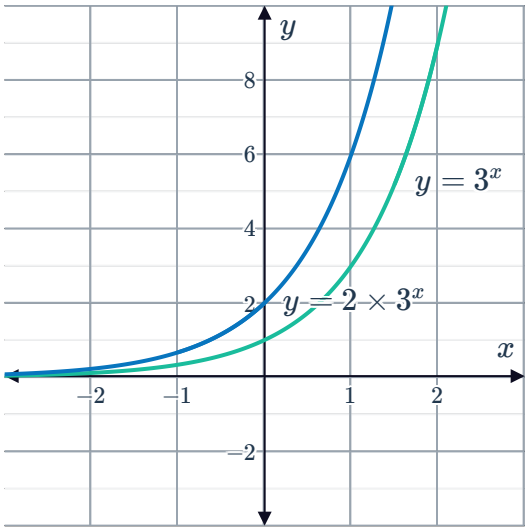
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a

Point on original graph	$\left(-1, \frac{1}{3}\right)$	$(0,1)$	$(1,3)$	$(2,9)$
Point on new graph	$\left(-1, \frac{2}{3}\right)$	$(0,2)$	$(1,6)$	$(2,18)$

b $y = 2 \times 3^x$

c



d Above

e Above

12

a Decreasing **b** 0 **c** $-\infty$ **d** $(0, -2.5)$

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